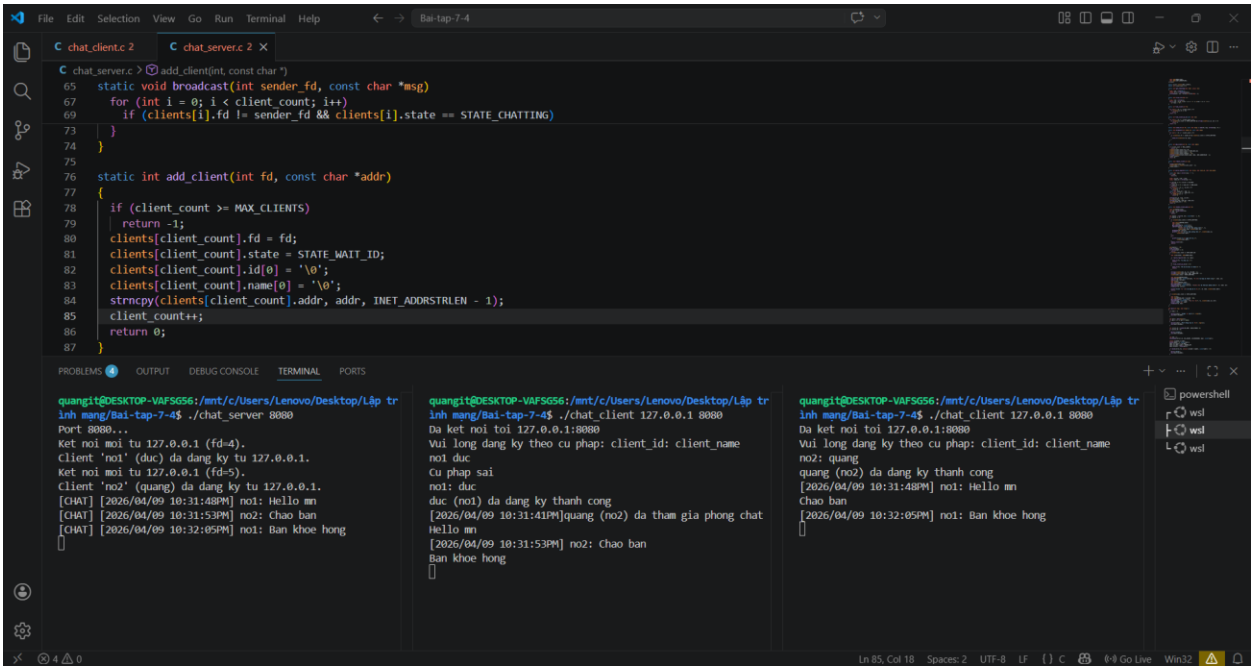
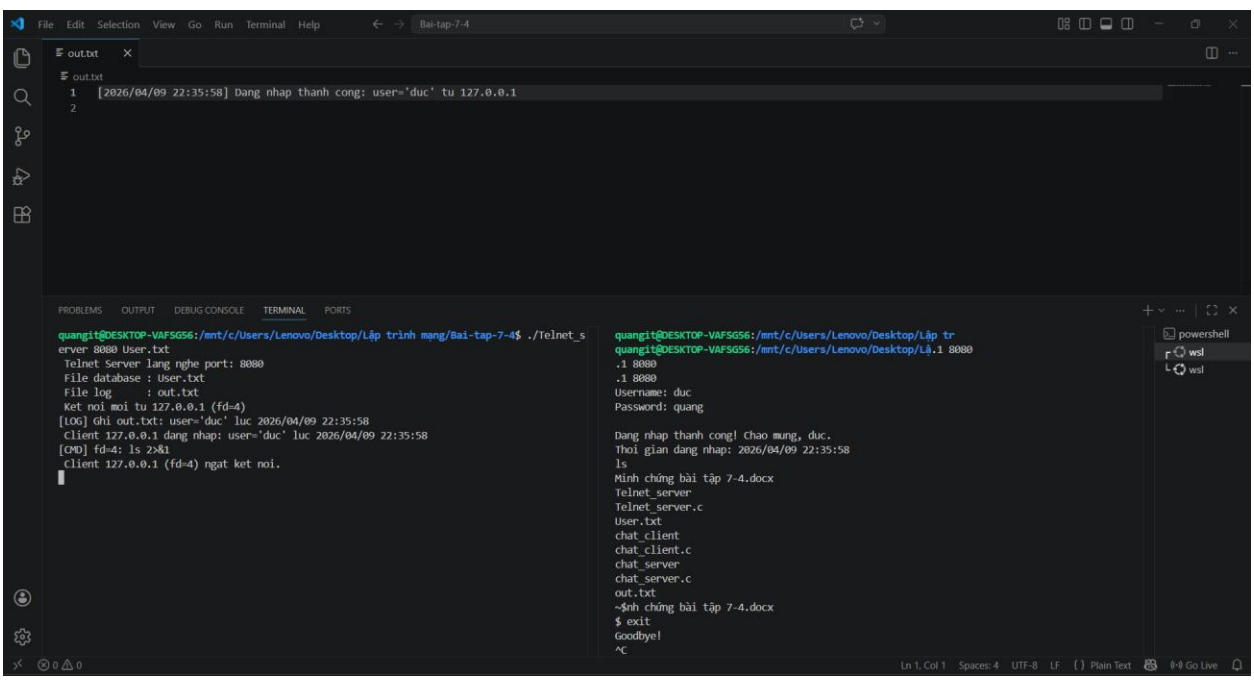
Bài 1:


chat_client.c 2
chat_server.c 2
chat_server.c > add_client(int, const char *)
static void broadcast(int sender_fd, const char *msg)
for (int i = 0; i < client_count; i++)
if (clients[i].fd != sender_fd && clients[i].state == STATE_CHATTING)
}
}
static int add_client(int fd, const char *addr)
{
if (client_count >= MAX_CLIENTS)
return -1;
clients[client_count].fd = fd;
clients[client_count].state = STATE_WAIT_ID;
clients[client_count].id[0] = '\0';
clients[client_count].name[0] = '\0';
strncpy(clients[client_count].addr, addr, INET_ADDRSTRLEN - 1);
client_count++;
return 0;
}
quangit@DESKTOP-VAFSG56:/mnt/c/Users/Lenovo/Desktop/Lập trình mạng/Bai-tap-7-4$ ./chat_server 8080
Port 8080...
Ket noi moi tu 127.0.0.1 (fd=4).
Client 'no1' (duc) da dang ky tu 127.0.0.1.
Ket noi moi tu 127.0.0.1 (fd=5).
Client 'no2' (quang) da dang ky tu 127.0.0.1.
[CHAT] [2026/04/09 10:31:48PM] no1: Hello mn
[CHAT] [2026/04/09 10:31:53PM] no2: Chao ban
[CHAT] [2026/04/09 10:32:05PM] no1: Ban khoe hong
quangit@DESKTOP-VAFSG56:/mnt/c/Users/Lenovo/Desktop/Lập trình mạng/Bai-tap-7-4$ ./chat_client 127.0.0.1 8080
Da ket noi toi 127.0.0.1:8080
Vui long dang ky theo cu phap: client_id: client_name
no1 duc
Cu phap sai
no1: duc
duc (no1) da dang ky thanh cong
[2026/04/09 10:31:41PM]quang (no2) da tham gia phong chat
Hello mn
[2026/04/09 10:31:53PM] no2: Chao ban
Ban khoe hong
quangit@DESKTOP-VAFSG56:/mnt/c/Users/Lenovo/Desktop/Lập trình mạng/Bai-tap-7-4$ ./chat_client 127.0.0.1 8080
Da ket noi toi 127.0.0.1:8080
Vui long dang ky theo cu phap: client_id: client_name
no2: quang
quang (no2) da dang ky thanh cong
[2026/04/09 10:31:48PM] no1: Hello mn
Chao ban
[2026/04/09 10:32:05PM] no1: Ban khoe hong

Bài 2:


out.txt
[2026/04/09 22:35:58] Dang nhap thanh cong: user='duc' tu 127.0.0.1
quangit@DESKTOP-VAFSG56:/mnt/c/Users/Lenovo/Desktop/Lập trình mạng/Bai-tap-7-4$ ./Telnet_server 8080 User.txt
Telnet Server lang nghe port: 8080
File database : User.txt
File log : out.txt
Ket noi moi tu 127.0.0.1 (fd=4)
[LOG] Ghi out.txt: user='duc' luc 2026/04/09 22:35:58
Client 127.0.0.1 dang nhap: user='duc' luc 2026/04/09 22:35:58
[CMD] fd=4: ls 2>&1
Client 127.0.0.1 (fd=4) ngat ket noi.
quangit@DESKTOP-VAFSG56:/mnt/c/Users/Lenovo/Desktop/Lập tr
quangit@DESKTOP-VAFSG56:/mnt/c/Users/Lenovo/Desktop/Lập.1 8080
.1 8080
Username: duc
Password: quang
Dang nhap thanh cong! Chao mung, duc.
Thoi gian dang nhap: 2026/04/09 22:35:58
ls
Minh chứng bài tập 7-4.docx
Telnet_server
Telnet_server.c
User.txt
chat_client
chat_client.c
chat_server
chat_server.c
out.txt
~$nh chứng bài tập 7-4.docx
$ exit
Goodbye!
^C